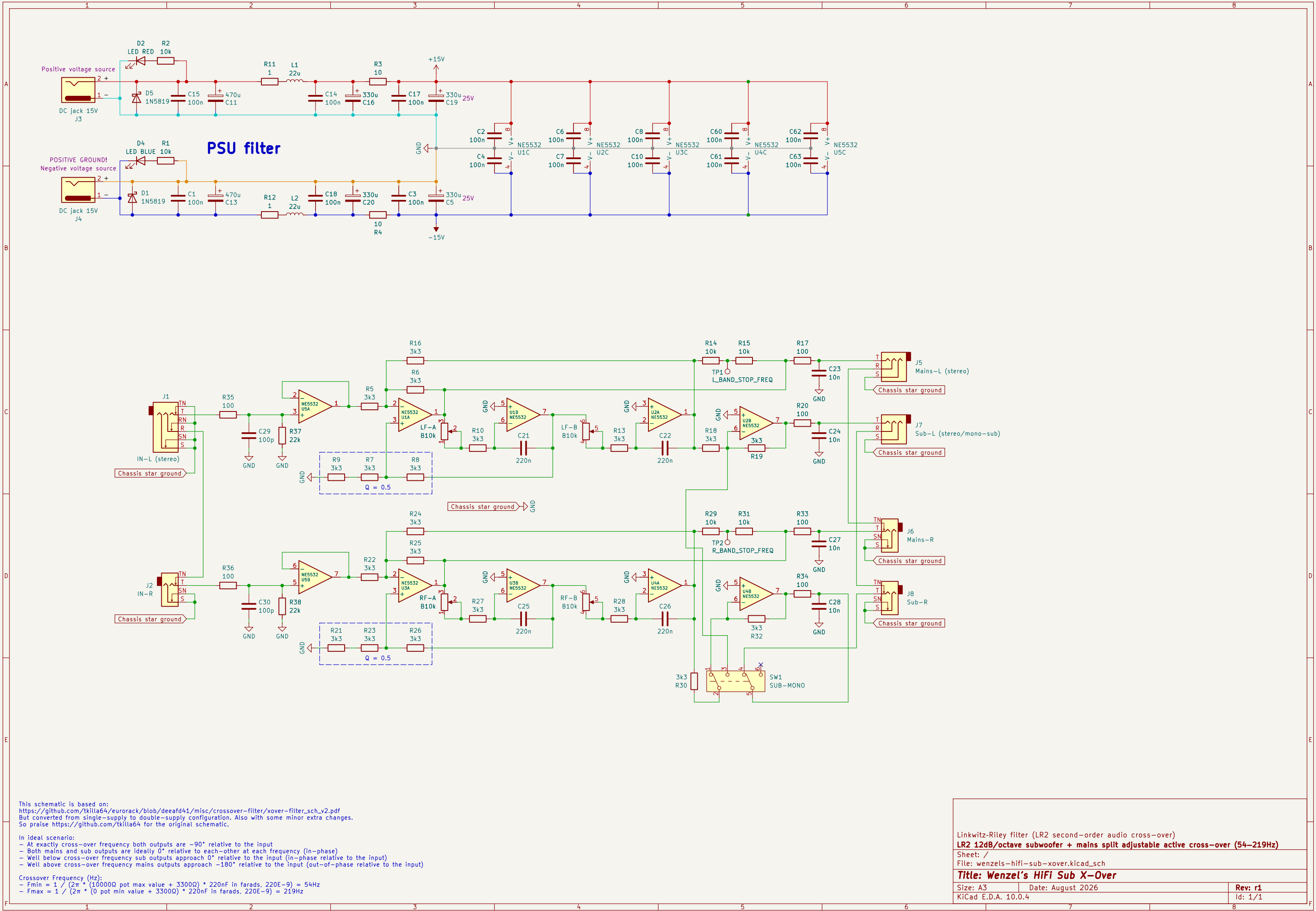




This schematic is based on:
https://github.com/tkill64/aurorack/blob/deaf41/misc/crossover-filter/xover-filter_sch.v2.pdf
But converted from single-supply to double-supply configuration. Also with some minor extra changes.
So praise <https://github.com/tkill64> for the original schematic.

- In ideal scenario:
- At exactly cross-over frequency both outputs are -90° relative to the input
 - Both mains and sub outputs are ideally 0° relative to each-other at each frequency (in-phase)
 - Well below cross-over frequency sub outputs approach 0° relative to the input (in-phase relative to the input)
 - Well above cross-over frequency mains outputs approach -180° relative to the input (out-of-phase relative to the input)

Crossover Frequency (Hz):

- $F_{min} = 1 / (2\pi * (10000\Omega \text{ pot max value} + 3300\Omega) * 220\text{nF in farads}, 220\text{E}-9) = 54\text{Hz}$
- $F_{max} = 1 / (2\pi * (0 \text{ pot min value} + 3300\Omega) * 220\text{nF in farads}, 220\text{E}-9) = 219\text{Hz}$

Linkwitz-Riley filter (LR2 second-order audio cross-over)
LR2 12dB/octave subwoofer + mains split adjustable active cross-over (54-219Hz)

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File: wenzels-hifi-sub-xover.kicad_sch

Title: Wenzel's HiFi Sub X-Over

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